

Explicit Body and Hand Pose for Social-Norm Reasoning

Idea 5: exploratory analysis draft for author review

October 4, 2026

Review status. The computational work below is complete. Human assessment of the detections and the final interpretation are pending. This standalone preview is not the team’s complete ICML submission; the section source is designed to be inserted into the existing report.

1 Analysis: Availability of Body and Hand Pose in EgoNormia

Motivation and questions. Idea 5 proposes adding explicit geometry and body/hand representations to RGB inputs for social-norm reasoning. We examine two premises: whether pretrained extractors produce candidate body/hand representations often enough to supply this pathway, and whether aligned maps provide an immediate benefit to a frozen VLM beyond extra-image and mismatched-map controls. We use EgoNormia’s pre-action visual context [Rezaei et al., 2025]. This is an output-availability audit and a zero-shot pilot, not a test of pose accuracy or the proposed fine-tuning.

Data and protocol. We select 40 items from distinct source videos using a fixed SHA256 ordering before detector inference, choosing one item per source. Five frames per pre-action clip are sampled at 10%, 30%, 50%, 70%, and 90% of its length (200 frames total). We apply frozen MediaPipe Pose Landmarker Full and Hand Landmarker models in independent-image mode, allowing up to four bodies and four hands. Detection and presence thresholds are 0.5. A body output is retained if at least eight of its 33 landmarks lie within the image and have predicted visibility and presence at least 0.5; we repeat the retention rule at 0.8 as a sensitivity check. The eight-point cutoff is an operational filter, not a validated measure of quality. Hand output means at least one predicted hand; the model does not supply equivalent per-landmark scores. We retain all examples, including blank outputs. Intervals use 5,000 source-video bootstrap samples.

Results. Body output satisfies the 0.5 retention rule in 59/200 sampled frames (29.5%) and the 0.8 rule in 59/200 (29.5%). At least one hand is predicted in 36/200 frames (18.0%). At least one body or hand output is available in 81/200 frames (40.5%); 31/40 clips contain

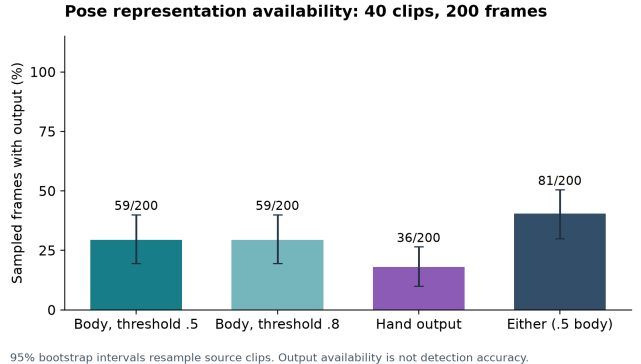


Figure 1: Availability of body and hand outputs on the fixed sample. Error bars are 95% bootstrap intervals clustered by source video. Body outputs require at least eight retained landmarks; predicted hand outputs use the detector’s standard acceptance threshold.

such an output in at least one sampled frame. These are detector-output frequencies, not precision, recall, or human-validated extraction quality.

Frozen-model pilot. On the same fixed items, we compare a local community 4-bit Qwen3-VL-8B-Instruct checkpoint under RGB-only, RGB plus repeated RGB, RGB plus aligned body/hand skeletons, and RGB plus skeletons from a different source video. RGB grids contain the same five chronological frames, each resized to a maximum side of 384 pixels; auxiliary grids match target grid dimensions. Repeated RGB controls for the extra image slot, and cyclically shuffled maps test alignment. Greedy decoding separately selects an action and then its justification. Action and justification options are independently permuted per item and kept fixed across conditions; no gold labels, norm labels, or generated descriptions enter prompts. Invalid outputs count as wrong. This is a custom subset pilot, not an official leaderboard replication.

Input	Action	Justification	Both
RGB	18/40	19/40	16/40
RGB + repeat	16/40	15/40	13/40
RGB + aligned pose	18/40	18/40	15/40
RGB + shuffled pose	18/40	19/40	15/40

Aligned pose minus RGB joint accuracy is -2.5 percentage points (95% paired bootstrap interval [-7.5, +0.0]);

0 items improve and 1 worsen. Against repeated RGB, the joint difference is +5.0 points [+0.0, +12.5]. Against shuffled pose, no item’s joint correctness changes. The resulting zero-width empirical bootstrap interval is degenerate and does not establish a zero population effect.

Discussion of this analysis (provisional; author review required). Aligned pose produces no observed joint-accuracy gain over RGB on this sample, and aligned and shuffled pose have identical joint-correctness outcomes on all 40 items. This provides no evidence of an alignment-dependent benefit under the tested setup. Repeating RGB also changes outcomes, demonstrating why an extra-image control matters. These results do not disprove the fine-tuning proposal: the frozen model has not been adapted to the rendered maps, and the sample is small. Candidate body or hand outputs are available in only 40.5% of sampled frames; human review must distinguish absent people/hands from detector failures or task-irrelevant detections. Egocentric views can show the wearer’s hands without a full body, so both extractors need separate assessment. Five independent frames do not measure motion or identity continuity. Before committing to pose augmentation, manually audit overlays, preserve missing-output indicators, and evaluate adaptation under matched budgets. EgoNormia is excluded from training but has been exposed during exploratory analysis; it should not be described as untouched.

References

MohammadHossein Rezaei, Yicheng Fu, Phil Cuvin, Caleb Ziems, Yanzhe Zhang, Hao Zhu, and Diyi Yang. Egonormia: Benchmarking physical-social norm understanding. In *Findings of the Association for Computational Linguistics: ACL 2025*, 2025. doi: 10.18653/v1/2025.findings-acl.985. URL <https://aclanthology.org/2025.findings-acl.985/>.

AI usage and contribution record (draft)

An OpenAI Codex agent inspected the assignment and proposal, suggested analysis approaches, selected the fixed sampling and control protocol, wrote and debugged the download/extraction/inference/reporting code, created isolated Python environments, downloaded the public dataset and pretrained models, executed the computations, calculated the summary statistics and intervals, and generated plots, an interactive review gallery, and this provisional report text. AI assistance therefore includes experimental design, code, execution support, analysis summaries, and writing; it must not be described as wording assistance alone.

Frozen MediaPipe Pose Landmarker Full and Hand Landmarker models generated body and hand landmark predictions. A locally executed community 4-bit Qwen3-VL-8B-Instruct checkpoint generated experimental action and justification choices. No model was fine-tuned. The agent did not supply human benchmark answers or human annotations.

The student selected research idea 5 and requested this exploratory work. No personal review of the generated overlays, final author interpretation, TA approval, or teammate contributions has yet been recorded. Add these details truthfully once completed. The assignment requires students to take a leading role in the project and disclose all AI use.

Files and reproducibility

The source-video sampling rule, frame timestamps, raw landmarks, frozen-model prompts and responses, option permutations, model revisions and hashes, dependency versions, download verification, and per-item results are saved in the accompanying project. The review gallery allows original frames, overlays, and skeleton maps to be inspected, with locally stored review notes exportable as JSON. All blank detections and invalid model responses remain in the denominator.

Checks before submission

Inspect the examples and identify mistaken or task-irrelevant detections. Replace the provisional discussion with your own conclusions. Confirm the analysis scope with the primary TA and incorporate the section and figures into the team's ICML report within its shared page limit. Complete the team-level disclosure and contribution statement.